

Assignment 4: Web Service Implementation

PRD & AI-Assisted Development Report

PromptCrafter - AI Prompt Builder

June 2, 2026

Part 1: Product Requirements Document (PRD)

1. Product Overview

PromptCrafter is an AI-assisted web service designed to help users generate highly optimized, professional-grade prompts for Large Language Models (LLMs) like ChatGPT, Claude, and Gemini. It solves the “blank canvas problem” by providing a structured, intuitive interface where users can select frameworks, roles, and magic keywords without needing to memorize prompt engineering techniques.

2. Target Users

- **Students & Academics:** Needs help structuring essays, summarizing papers, or planning study schedules.
- **Developers:** Needs help debugging code, writing documentation, or understanding new frameworks.
- **Marketers & Creators:** Needs engaging copy, content strategies, and SEO optimization.
- **General Public (Beginners):** People who want to use AI effectively but don't know how to write good prompts.

Pain Points Addressed: Low quality AI responses due to vague instructions, lack of knowledge about prompt engineering, and the repetitive nature of typing out constraints.

3. Project Goals

- Provide a 1-minute, zero-learning-curve prompt generation experience.
- Implement at least 4 proven prompt frameworks (Basic, Chain of Thought, Persona, Extraction).
- Ensure the output is instantly usable via a 1-click copy feature.
- Maintain a visually pleasing, modern glassmorphism UI.

4. Core User Scenario

1. A user wants to write a marketing email but doesn't know how to instruct the AI.
2. The user opens PromptCrafter.
3. They select the “Deep Persona” framework and choose the “Digital Marketer” role.
4. They enter their task: “Write a launch email for a new productivity app.”
5. They open “Advanced Options” and check the “Virtual Tip” and “Reverse Questioning” magic keywords.
6. A perfectly structured markdown prompt is dynamically generated.
7. The user clicks “Copy” and pastes it into ChatGPT.
8. The user clicks “Save to Library” to reuse this setup later.

5. Feature List

- **Framework Selection** (Must-have): Dropdown to select different prompt engineering structures (RTF, CoT, etc.).
- **Dynamic Prompt Generator** (Must-have): Real-time text generation based on inputs.
- **Advanced Options** (Should-have): Inputs for Context, Target Audience, and Constraints.
- **Prompt Boosters / Magic Keywords** (Should-have): Checkboxes to inject proven prompt hacks (e.g., Virtual Tip, Bias Prevention).
- **Clipboard Copy** (Must-have): 1-click copy functionality.
- **My Library** (Nice-to-have): Save and delete favorite prompts using Local Storage.

6. Page Structure

The application is a Single Page Application (SPA) divided into three logical sections:

1. **Home Section:** Landing area with a hero title, brief description, and a “Start” button.
2. **Builder Section:** The core tool area containing dropdowns, inputs, advanced options toggle, and the real-time result preview.
3. **Library Section:** A dashboard grid displaying previously saved prompts.

7. Technical Requirements

- **Frontend:** HTML5, Vanilla CSS3, Vanilla JavaScript (ES6+). No frontend frameworks (React/Vue) are used to keep it lightweight.
- **Styling:** CSS variables for theming, CSS Flexbox/Grid for layout, and backdrop-filter for glassmorphism.
- **Storage:** Browser `localStorage` API for the Library feature (No backend/database).
- **Deployment:** Vercel (Static Site Deployment).

8. Design Requirements

- **Theme:** Dark Mode by default.
- **Aesthetics:** “Glassmorphism” - semi-transparent panels with background blur over floating, animated gradient orbs.
- **Typography:** ‘Outfit’ Google Font for a modern, geometric, and clean look.
- **Interactions:** Smooth CSS transitions for button hovers, section toggling, and expanding the advanced options panel.

9. Milestones

1. **Idea Definition & PRD Completion:** Outline the problem, solution, and core features.
2. **First Prototype:** Implement the basic HTML/CSS skeleton and simple string-concatenation JS logic.
3. **Advanced Implementation:** Integrate prompt frameworks and magic keywords based on prompt engineering research.
4. **Refinement & Testing:** Add smooth UI transitions and test Local Storage persistence.
5. **Final Deployment:** Push code to GitHub and deploy publicly via Vercel.

Part 2: AI-Assisted Development Report

1. AI Tools Used

- Gemini 3.1 Pro (via Antigravity IDE)

2. Tasks Assisted by AI

- PRD Writing: I came up with the core idea for a prompt generator and used the AI to help structure my thoughts into a professional Product Requirements Document (PRD) that meets the assignment guidelines.
- UI/UX Design & Publishing: I asked the AI to help write the HTML skeleton and apply modern CSS styling, including glassmorphism, dark mode, and background animations.
- JavaScript Implementation: I used the AI to help implement the core logic, such as real-time prompt assembly, clipboard copying, and saving data using the browser Local Storage.
- Prompt Engineering Logic: I researched prompt engineering techniques (like Chain of Thought and reverse questioning) and instructed the AI to implement these dynamic frameworks and magic keyword checkboxes.

3. Representative Prompts Used

1. I have an idea for a custom AI prompt generator. Help me draft a professional PRD that includes the Product Overview, Target Users, and Feature List.
2. Based on the PRD we wrote, please generate the index.html, style.css, and script.js files. Make sure the design uses a modern glassmorphism aesthetic and a dark theme.
3. I read an article about prompt hacking techniques like promising a tip and asking for reverse questions. Please add these as magic keyword checkboxes in our web application.

4. Modifications and Improvements

- Natural Prompt Generation: The initial string concatenation method suggested by the AI was too basic. I completely overhauled the engine to use a switch-case statement that handles 4 distinct frameworks (Basic, Chain of Thought, Persona, Extraction) and outputs structured markdown.
- Prompt Boosters: Based on my own research, I added logic and UI for toggleable advanced options. These magic keywords (e.g., promising a virtual tip, inducing reverse questions) are automatically appended to the bottom of the prompt when checked, significantly improving the output quality.
- Design Details: The initial background gradient animation was too fast and distracting. I adjusted the CSS animation duration to 10 seconds to create a much smoother and less intrusive visual experience, and added smooth dropdown animations to the advanced options panel.

5. Bugs, Errors, and Limitations Found

- Issue: When opening the browser to use the library (Local Storage) feature for the first time, the application threw a JavaScript error because the stored data was null, which broke the array forEach method.
- Solution: I fixed this by adding a logical OR operator when parsing the JSON data. If the data is empty or null, it now returns an empty array as the default value, successfully resolving the error.

6. Final Vercel Deployment URL

- <https://promptcrafter-six.vercel.app>